GazeVoice

GazeVoice Chess — An AI-powered desktop chess app combining voice and webcam-based gaze control to improve accessibility for players with limited hand or motor control.

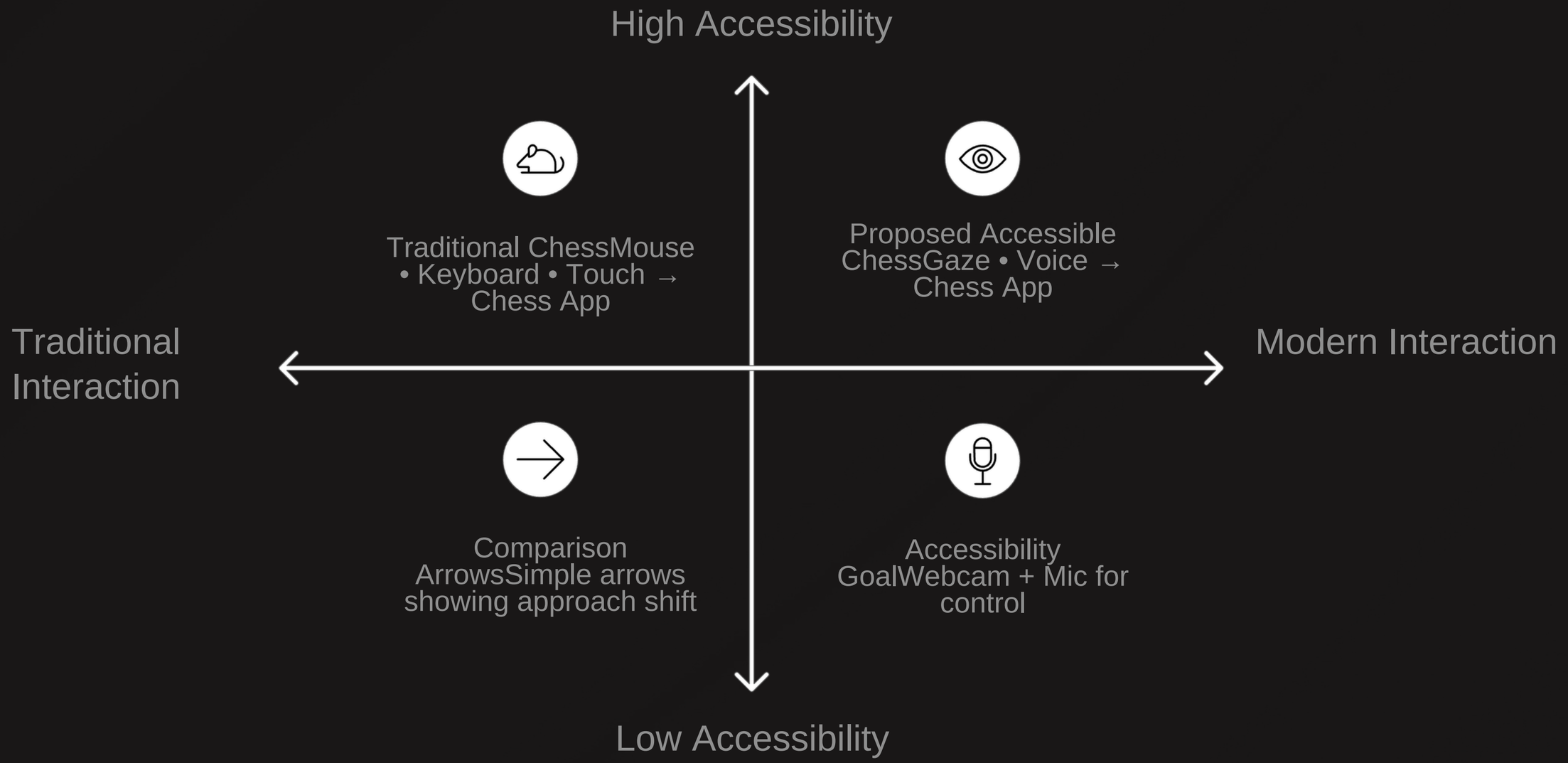
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- CSE

- Foundations of Artificial Intelligence



Problem Statement

Traditional chess interfaces rely on mouse, keyboard, or touch — barriers for users with limited motor control. Specialized eye-trackers are costly or unavailable. We investigate whether a standard webcam + microphone can enable usable gaze and voice control for chess.



Proposed Solution

GazeVoice Chess — a Python desktop prototype focused on accessible interaction modes and understandable AI for education.

Key Features

- Human vs AI gameplay
- Voice-based chess commands
- Webcam-based gaze interaction
- Legal move validation (python-chess)
- Multiple AI difficulty levels
- Voice feedback and accessibility settings



Technology Stack

Compact reference of technologies used and their roles.

Technology	Purpose
Python	Main programming language
Tkinter	Desktop graphical interface
python-chess	Chess rules, legal move validation
Minimax + Alpha-Beta	AI decision-making and pruning
OpenCV	Webcam capture and image preprocessing
MediaPipe	Facial landmarks for gaze estimation
SpeechRecognition	Voice command capture
pyttsx3	Local voice feedback
VS Code + AI Assistant	Development and iterative coding support

AI Implementation — Minimax

Design goal: an educational, understandable AI rather than a world-class engine.



Key Points

- Minimax evaluates future positions considering opponent responses
- Alpha-beta pruning reduces examined branches
- Evaluation function scores material + simple positional factors
- Search depth adjustable by difficulty: Easy (low) • Medium (moderate) • Hard (higher)

System Architecture

Technical architecture showing data flow between user, modules, and outputs.

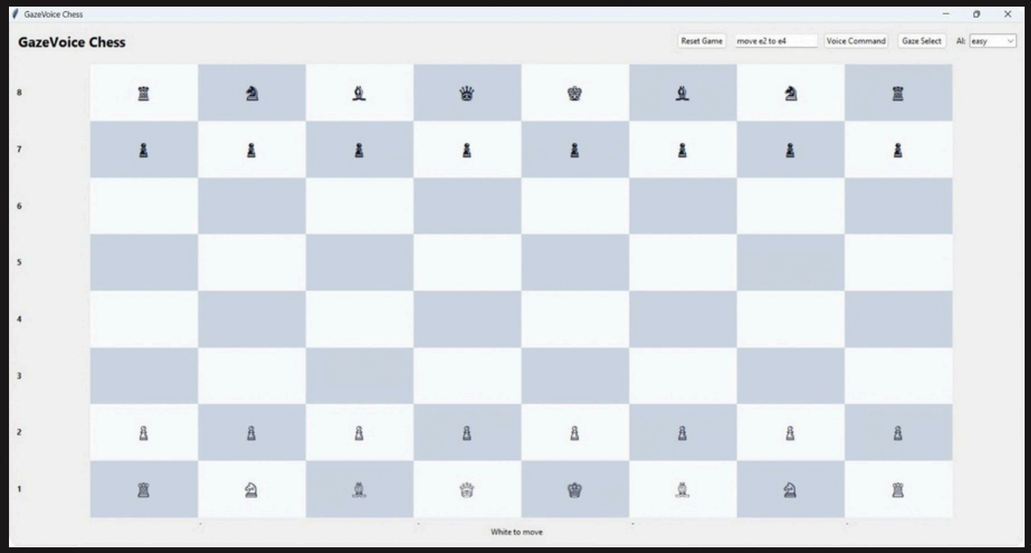


Current Progress & Demo

Current Progress

- ✓ Project structure created
- ✓ Chessboard interface
- ✓ Legal move handling (python-chess)
- ✓ Basic AI opponent implemented
- ✓ AI difficulty levels exposed
- ✓ Accessibility interface prototype (UI hooks)
- ◐ Webcam gaze tracking — prototype
- ◐ Voice interaction — prototype
- ◌ Hardware testing — pending

Demo Placeholders



Limitations & Future Scope

Current Limitations

- Webcam gaze tracking not fully implemented
- Extensive camera/environment testing still required
- Voice interaction is prototype-level; error handling limited
- AI is intentionally simple for education (not competitive)
- End-to-end accessibility and usability testing pending
- No formal compliance or assistive-device certification yet

Future Scope

- Complete gaze calibration and improve estimation robustness
- Integrate noise-robust live microphone recognition
- Refine evaluation function and tune search heuristics
- Optional integration with stronger chess engines (as needed)
- Conduct usability studies with assistive users and iterate
- Expose more accessibility customization (timing, confirmation, UI scale)

Thank You!